

## SAFETY DATA SHEET

Creation Date 16-November-2010

Revision Date 28-March-2024

Revision Number 3

### 1. Identification

**Product Name** Lithium aluminum hydride

**Cat No. :** A18116

**CAS-No** 16853-85-3  
**Synonyms** LAH; Lithium tetrahydridoaluminate

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                                        |              |                    |
|------------------------------------------------------------------------|--------------|--------------------|
| Substances/mixtures which, in contact with water, emit flammable gases | Category 1   | Gas(es) = Hydrogen |
| Skin Corrosion/Irritation                                              | Category 1 A |                    |
| Serious Eye Damage/Eye Irritation                                      | Category 1   |                    |
| Combustible Dusts                                                      | Category 1   |                    |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

May form combustible dust concentrations in air

In contact with water releases flammable gases which may ignite spontaneously

Causes severe skin burns and eye damage



### Precautionary Statements

#### Prevention

Keep container tightly closed

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking

Do not allow contact with water

Do not breathe dust/fumes/gas/mist/vapours/spray

Wash face, hands and any exposed skin thoroughly after handling

Wear protective gloves/protective clothing/eye protection/face protection

Handle and store contents under inert gas. Protect from moisture

#### Response

In case of major fire and large quantities: Evacuate area. Fight fire remotely due to the risk of explosion

IF SWALLOWED: Rinse mouth. Do NOT induce vomiting

IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower

IF INHALED: Remove person to fresh air and keep comfortable for breathing

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

Immediately call a POISON CENTER/doctor

Wash contaminated clothing before reuse

In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish

IF ON SKIN: Brush off loose particles from skin. Immerse in cool water

#### Storage

Store locked up

Store in a dry place. Store in a closed container

Store in a well-ventilated place. Keep container tightly closed

#### Disposal

Dispose of contents/container to an approved waste disposal plant

## 3. Composition/Information on Ingredients

| Component                | CAS-No     | Weight % |
|--------------------------|------------|----------|
| Lithium aluminum hydride | 16853-85-3 | 95       |

## 4. First-aid measures

|                                        |                                                                                                                                                                               |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Eye Contact</b>                     | Immediate medical attention is required. Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes.                                             |
| <b>Skin Contact</b>                    | Wash off immediately with soap and plenty of water while removing all contaminated clothes and shoes. Immediate medical attention is required.                                |
| <b>Inhalation</b>                      | Remove from exposure, lie down. Remove to fresh air. If not breathing, give artificial respiration. Immediate medical attention is required.                                  |
| <b>Ingestion</b>                       | Do NOT induce vomiting. Never give anything by mouth to an unconscious person. Drink plenty of water. Call a physician immediately.                                           |
| <b>Most important symptoms/effects</b> | Causes burns by all exposure routes. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should |

## Notes to Physician

be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation  
Treat symptomatically

## 5. Fire-fighting measures

|                                         |                          |
|-----------------------------------------|--------------------------|
| <b>Suitable Extinguishing Media</b>     | Dry chemical.            |
| <b>Unsuitable Extinguishing Media</b>   | No information available |
| <b>Flash Point</b>                      | No information available |
| <b>Method -</b>                         | No information available |
| <b>Autoignition Temperature</b>         | Not applicable           |
| <b>Explosion Limits</b>                 |                          |
| <b>Upper</b>                            | No data available        |
| <b>Lower</b>                            | No data available        |
| <b>Sensitivity to Mechanical Impact</b> | No information available |
| <b>Sensitivity to Static Discharge</b>  | No information available |

### Specific Hazards Arising from the Chemical

Dust can form an explosive mixture with air. Contact with water liberates toxic gas. Water reactive. Combustible material. Produce flammable gases on contact with water. Fine dust dispersed in air may ignite.

### Hazardous Combustion Products

Hydrogen. Burning produces obnoxious and toxic fumes.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

**Health**  
3

**Flammability**  
2

**Instability**  
2

**Physical hazards**  
W

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                                                 |
|---------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required.                                                                                                                                                                     |
| <b>Environmental Precautions</b>            | Should not be released into the environment.                                                                                                                                                                                                    |
| <b>Methods for Containment and Clean Up</b> | Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment. Sweep up and shovel into suitable containers for disposal. Keep from any possible contact with water, because of violent reaction and possible flash fire. |

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                                                                                                                                                                    |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Do not breathe dust. Do not get in eyes, on skin, or on clothing. Handle product only in closed system or provide appropriate exhaust ventilation. Use spark-proof tools and explosion-proof equipment. Use only non-sparking tools. Do not allow contact with water.                                                                                                              |
| <b>Storage.</b> | Keep in a dry, cool and well-ventilated place. Protect from direct sunlight. Keep from any possible contact with water. Keep under nitrogen. Keep away from water or moist air. Keep at temperatures below 35°C. Keep containers tightly closed in a dry, cool and well-ventilated place. Corrosives area. Incompatible Materials. Acids. Water. Alcohols. Strong reducing agents. |

## 8. Exposure controls / personal protection

|                            |                                                                                                                                               |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Guidelines</b> | This product does not contain any hazardous materials with occupational exposure limits established by the region specific regulatory bodies. |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|

**Engineering Measures**

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

**Personal protective equipment**

**Eye Protection**  
**Hand Protection**

Goggles  
Protective gloves

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Natural rubber | See manufacturers | -               | Splash protection only |
| Nitrile rubber | recommendations   |                 |                        |
| Neoprene       |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

**Environmental exposure controls**

No information available.

**Hygiene Measures**

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                         |                          |
|-----------------------------------------|--------------------------|
| <b>Physical State</b>                   | Powder Solid             |
| <b>Appearance</b>                       | Off-white                |
| <b>Odor</b>                             | No information available |
| <b>Odor Threshold</b>                   | No information available |
| <b>pH</b>                               | No information available |
| <b>Melting Point/Range</b>              | 125 °C / 257 °F          |
| <b>Boiling Point/Range</b>              | No information available |
| <b>Flash Point</b>                      | No information available |
| <b>Evaporation Rate</b>                 | Not applicable           |
| <b>Flammability (solid,gas)</b>         | No information available |
| <b>Flammability or explosive limits</b> |                          |
| Upper                                   | No data available        |
| Lower                                   | No data available        |
| <b>Vapor Pressure</b>                   | negligible               |
| <b>Vapor Density</b>                    | Not applicable           |
| <b>Specific Gravity</b>                 | 0.910                    |

|                                        |                          |
|----------------------------------------|--------------------------|
| Solubility                             | No information available |
| Partition coefficient; n-octanol/water | No data available        |
| Autoignition Temperature               | Not applicable           |
| Decomposition Temperature              | 125 °C                   |
| Viscosity                              | Not applicable           |
| Molecular Formula                      | Al H4 Li                 |
| Molecular Weight                       | 37.95                    |

## 10. Stability and reactivity

|                                  |                                                                                                                                                          |
|----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Reactive Hazard                  | Yes                                                                                                                                                      |
| Stability                        | Risk of explosion by shock, friction, fire or other sources of ignition. heat sensitive. Water reactive.                                                 |
| Conditions to Avoid              | Keep away from open flames, hot surfaces and sources of ignition. Excess heat. Exposure to light. Incompatible products. Exposure to moist air or water. |
| Incompatible Materials           | Acids, Water, Alcohols, Strong reducing agents                                                                                                           |
| Hazardous Decomposition Products | Hydrogen, Burning produces obnoxious and toxic fumes                                                                                                     |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                                                                 |
| Hazardous Reactions              | Contact with water liberates extremely flammable gases.                                                                                                  |

## 11. Toxicological information

### Acute Toxicity

|                                      |                                                             |
|--------------------------------------|-------------------------------------------------------------|
| Product Information                  | No acute toxicity information is available for this product |
| Component Information                |                                                             |
| Toxicologically Synergistic Products | No information available                                    |

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | Causes burns by all exposure routes                                                      |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                | CAS-No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|--------------------------|------------|------------|------------|------------|------------|------------|
| Lithium aluminum hydride | 16853-85-3 | Not listed | Not listed | Not listed | Not listed | Not listed |

|                                            |                                                                                                                                                                                                                                                            |
|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Mutagenic Effects                          | No information available                                                                                                                                                                                                                                   |
| Reproductive Effects                       | No information available.                                                                                                                                                                                                                                  |
| Developmental Effects                      | No information available.                                                                                                                                                                                                                                  |
| Teratogenicity                             | No information available.                                                                                                                                                                                                                                  |
| STOT - single exposure                     | None known                                                                                                                                                                                                                                                 |
| STOT - repeated exposure                   | None known                                                                                                                                                                                                                                                 |
| Aspiration hazard                          | No information available                                                                                                                                                                                                                                   |
| Symptoms / effects, both acute and delayed | Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Reacts with water so no ecotoxicity data for the substance is available.

**Persistence and Degradability** Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No UN1410  
 Proper Shipping Name LITHIUM ALUMINUM HYDRIDE  
 Hazard Class 4.3  
 Packing Group I

### TDG

UN-No UN1410  
 Proper Shipping Name LITHIUM ALUMINUM HYDRIDE  
 Hazard Class 4.3  
 Packing Group I

### IATA

UN-No UN1410  
 Proper Shipping Name LITHIUM ALUMINIUM HYDRIDE  
 Hazard Class 4.3  
 Packing Group I

### IMDG/IMO

UN-No UN1410  
 Proper Shipping Name LITHIUM ALUMINIUM HYDRIDE  
 Hazard Class 4.3  
 Packing Group I

## 15. Regulatory information

### International Inventories

| Component                | CAS-No     | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|--------------------------|------------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Lithium aluminum hydride | 16853-85-3 | -   | X    | X    | ACTIVE                                        | 240-877-9 | -      | -   |

| Component                | CAS-No     | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|--------------------------|------------|-------|----------|------|------|------|------|-------|-------|
| Lithium aluminum hydride | 16853-85-3 | X     | KE-00990 | X    | X    | X    | X    | X     | X     |

### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances  
KECL - Korean Existing and Evaluated Chemical Substances  
ENCS - Japanese Existing and New Chemical Substances  
AICS - Australian Inventory of Chemical Substances  
PICCS - Philippines Inventory of Chemicals and Chemical Substances

#### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

#### Other International Regulations

##### Authorisation/Restrictions according to EU REACH

| Component                | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|--------------------------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Lithium aluminum hydride | -                                                                   | Use restricted. See item 75. (see link for restriction details)               | -                                                                                                     |

##### REACH links

<https://echa.europa.eu/substances-restricted-under-reach>

##### Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                | CAS-No     | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|--------------------------|------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| Lithium aluminum hydride | 16853-85-3 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                | CAS-No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|--------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Lithium aluminum hydride | 16853-85-3 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

##### Prepared By

Product Safety Department  
Email: [chem.techinfo@thermofisher.com](mailto:chem.techinfo@thermofisher.com)  
[www.thermofisher.com](http://www.thermofisher.com)

##### Creation Date

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##### Revision Date

28-March-2024

##### Print Date

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##### Revision Summary

New emergency telephone response service provider.

##### Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**